N2S Upstream Workbench — Complete Tour

Background · experiments · :8258 UI · code · claim hygiene

CXR N2S lab notes (port 8258)

September 2026

About this document

Complete Upstream write-up: background concepts, every experiment (U0→U-A→gen→causal-d→:8258), UI controls, and code — with a full decode of $d = \text{unit}(\mu_T - \mu_C)$. Twin Downstream tour (port 8257): N2S-Workbench-Newcomer-Tour.pdf. Lab-scale only — not clinical validation.

Contents

Part I — Background (keep these concepts clear)

1. The clinical question

We care about one formal predicate:

> Does the note satisfy **`FIRST_LINE_THERAPY_FAILED`**?

Notes can be:

Kind	Idea	Example shape
Temporal-change	Response / benefit **then later** progression	FOLFOX helps → months later grows → stop
Contradiction	Same-time conflict	"Failed" and "responsive" in one visit
Clear fail / nofail	Explicit progression or ongoing response	Straightforward E-notes

A common failure mode: the model marks **`contradiction.present = true`** **(X=true)** on a *temporal* note that is **not** a true same-time contradiction (**false X**).

2. Three pieces of the N2S system

```
Doctor's note
      ↓  UPSTREAM — formation of internal reps (THIS DOCUMENT · :8258)
Tokenization → layer-wise compute → residual stream
      ↓  DOWNSTREAM — utilization → structured LLM output (:8257)
Generation / commit (e.g. contradiction.present)
════════════════════════════
   NEURAL → SYMBOLIC BOUNDARY  (Track A · hard cut)
════════════════════════════
CXR atoms / Dual / gates → AUTO | REVIEW
```

Piece	Question	Port
Upstream	How does the representation **form** in prefill?	**8258**
Downstream	How is it **used** at commit / can we steer it?	**8257**
Boundary	Can we contain bad transforms as REVIEW?	**8257** Evaluate

Upstream vs downstream is an **experimental partition**, not a hard cut inside the transformer. The only hard product cut is neural → symbolic.

3. Prefill vs commit (must not mix)

Stage	When	What we measure
Prefill	Model reads the whole prompt (note + instructions) once	Residuals at note-body tokens × layers → **formation**
Commit	During generation, when writing `contradiction.present: true/false`	Logits / margin / X → **decision**

Upstream Score projects **prefill** activations onto **d**.
Downstream Intervene adds α·**v** at **commit**.
Same model; different question.

4. Activations, layers, residual stream

A transformer layer outputs a high-dimensional vector per token (hidden size ≈ 3584 for Qwen2.5-7B).

- **Layer Ln** = after block *n* (we use absolute indices: L8, L12, L16, L20, L24).
- **Residual stream** = the running representation those blocks write into.
- **Attn / MLP** = two writers inside a block (attention and feed-forward).

We capture these with HF **forward hooks** (callbacks on module outputs). We do **not** require TransformerLens for this work.

5. Directions in activation space (RepEng idea)

If a concept lives as a **direction**, you can:

1. **Read** — project activations onto that direction (Score / Deep dive)
2. **Write** — add a multiple of that direction (Intervene)
3. **Ablate** — zero or replace a component write (U-B, circuit C1)

Reading ≠ writing. A strong Score does **not** prove an editor.

6. Glossary — how to read `d = unit(μ_T − μ_C)`

This is the Upstream formation direction. Read it left to right:

Symbol	Plain English
T	Temporal-change discovery note (`EX_TEMPORAL_FOLFOX`) — response then later failure
C	Same-time contradiction discovery note (`EX_CONTRA`)
h	Residual vector at one token, one layer (length D)
μ_T	**Mean** of h over **all note-body tokens** of the temporal note at that layer
μ_C	Same mean for the contradiction note
μ_T − μ_C	Difference vector: points toward "more like temporal mean / less like contra mean"
unit(·)	Divide by Euclidean length → length **1**. Direction only; removes overall scale
d	That unit difference, **per layer** (so there is a d@L8, d@L12, … d@L24)

Bake (once, then freeze):

```
for each layer L:
    μ_T = mean over tokens of residual_T[L]
    μ_C = mean over tokens of residual_C[L]
    d[L] = (μ_T − μ_C) / ||μ_T − μ_C||
```

Cached as `artifacts/n2s-upstream-ua-directions.pt`.

Score a new note:

```
for each note-body token t at layer L:
    score = residual[t,L] · d[L]     # dot product
best layer ≈ argmax mean |score|
```

Quantity on UI	Meaning
|μ·d|	Mean absolute token·d at that layer (how strongly the note aligns with d)
Best layer	Layer with largest |μ·d| (usually **L24** for temporal-ish notes)
cos(d, v)	Angle between Upstream **d** and Downstream freeze-**v** (~0.03 → almost unrelated)
tok / score_d	Which token pieces light up along **d** (correlational hotspots)
cos_v	That token's cosine to Downstream **v** (usually near 0)

Forbidden reading: "These top tokens *are* the temporality circuit." They are **correlational** sites.

7. Glossary — Downstream sibling `v = unit(μ_A − μ_B)`

Do not confuse with **d**.

Symbol	Upstream **d**	Downstream **v**
Built from	Prefill note means (temporal vs contra)	Commit hiddens Class A vs B
Strongest layer	**L24**	**L20**
Used for	Formation Score / map	Steer / SAE / circuit
Causal editor?	**No** (ablate + α·d null)	**Partial** (α=8) + L20 MLP site

```
v = unit( mean(h_A) − mean(h_B) )   # expand fit @ L20 commit
h ← h + α · v                         # Intervene at contradiction.present
```

8. X, margin, soft gates

Term	Meaning
X	`contradiction.present` boolean in the JSON extract
Margin	logit(true) − logit(false) at the commit token
Soft gate	Pre-registered pass/fail rule for a panel (not fishing after the fact)
Null / weak	Soft gate failed — no honest causal claim

Part II — Everything we did (Upstream science)

Timeline at a glance

Step	Name	Result	Status
U0	Prefill cue × layer · freeze-v	Weak cue·v	Done
U1 / U1b	Prefill ±α·v at cue / L16 sites	**No X flip**	Done · stop α-chase
U-A	Multi-token class-mean **d** map	Strongest **L24**; cos(d,v)≈0.03	Done

Step	Name	Result	Status
U-B	Zero-ablate attn/mlp/resid @ L24 sites	**Null**	Done · paused
U-B2	Mean-ablate @ L20 sites	**Null**	Done · paused
U-A gen	Held-out paraphrases vs frozen **d**	**soft+strong YES** @ L24	**FROZEN**
U-A causal-d	Steer α·d @ L24 (commit + prefill)	**NULL/weak**	Done
`:8258`	Live Score / Frozen replay / Custom pair	Correlational UI	Live
U-C / U-D	Upstream SAE / circuits	**Not run** — locked off nulls	Locked

U0 — Prefill trace

Quest: Do freeze-**v** (commit editor) sit on obvious note cues during prefill?

Method: One prefill forward; cue-token residuals at layers {8,12,16,20,24}; project onto freeze-**v**.

Result: Weak. Commit-fitted **v** is **not** obviously "sitting on" recipe cues in prefill.

Artifact / doc: `TRACKB-UPSTREAM-PREFILL.md` · `n2s_upstream_prefill.py`

U1 / U1b — Prefill position steer

Quest: If we add ±α·v at cue positions on prefill, does commit X flip?

Method: `prefill_position_steer` in `hf_intervene.py`; α=8 at recipe / L16 outcome sites.

Result: **Null** — no X flip; margins flat. Contra stays true.

Decision: Stop α-chase on freeze-**v** × cue. Formation ≠ automatic commit editor.

U-A — Formation map (the core Upstream positive)

Quest: Where do temporal vs contradiction notes separate in prefill residual space?

Method:

1. Capture note-body residuals for discovery pair.
2. Per layer: `d = unit(μ_T − μ_C)`.
3. Rank layers by ‖μ_T − μ_C‖ / mean |token·d|.
4. List top tokens by |score|.

Result:

Finding	Detail
Best layer cos(d, freeze-v) Story	**L24** ≈ **0.03** Late-layer formation map; **not** the same object as Downstream **v**

Artifacts: `n2s-upstream-ua-map.json` · `n2s-upstream-ua-directions.pt` · `n2s-upstream-ua-readout.json`
CLI: `./scripts/run_upstream_ua.sh map`

Formation hypothesis (allowed):

Treatment-course vs contradiction geometry becomes visible late in prefill (esp. L24); it is correlationally distinct from the L20 commit editor **v**.

U-B / U-B2 — Ablation (causal test #1) → NULL

Quest: Are U-A top sites **necessary** for commit X?

Variant	Layer	Mode	Result
U-B	L24	Zero attn / mlp / resid at top sites	**No X flip**
U-B2	L20	Mean-ablate component writes	**No X flip**

Claim: Localization alone ≠ control. Ablation family **paused**.
Do not open U-C SAE / U-D from these nulls.

Artifacts: `n2s-upstream-ub-patch.json` · `n2s-upstream-ub2-patch.json` (+ readouts)

U-A gen — Paraphrase generalization → FROZEN YES

Quest: Does frozen **d** (baked only on discovery pair) separate **held-out** near/far paraphrases?

Gates:

Gate	Meaning	Result
Soft	mean_T > mean_C @ L24	**YES** (~58.4 > ~23.5)
Strong	min_T > max_C @ L24	**YES**

Claim: Correlational **lexical generalization** of the formation map.
Not a causal editor.

Freeze: `TRACKB-UPSTREAM-UA-GEN-FREEZE.md`
Data: `data/heldout-ua-paraphrase.json`

U-A causal-d — Steer along frozen d → NULL/weak

Quest: Does α·d edit X / margin dose-dependently?

Method: Commit + prefill steer @ L24; α ∈ {1,2,4,8}; ±d + gaussian; FOLFOX + CONTRA.

Result: **NULL/weak** both sites — no X flip; margin noise; contra held.
(Note: FOLFOX baseline already X=false on some runs — dull for false-X clear, still no dose editor.)

Distinct from U-B: direction steer vs top-site ablate; **both** null families.

Doc: `TRACKB-UPSTREAM-UA-CAUSAL.md`

Portfolio plain English

> I traced where a treatment-failure representation becomes visible inside a transformer, then tested whether the strongest locations (or the direction itself) controlled the model's decision. They did not. Localization alone was insufficient for causality. The map still **generalizes** to held-out paraphrases and lights up real temporal notes on :8258.

Part III — Workbench UI (:8258)

Run

```
cd cxr-n2s-eval-workbench
../cxrlabs/faiss_gpu1/bin/python upstream_server.py
# → http://127.0.0.1:8258/
```

Needs ~14 GiB free GPU for Score / pair.

Left panel — every control

Control	What it does	What it does **not**
Case id	Labels the JSON artifact	Change the model
Optional gold	Metadata only	Drive Score math
Clinical note	Input for **Score**	Used by Frozen replay
Score vs frozen d	Project this note onto frozen **d**	Give AUTO / SATISFIED
Frozen U-A/U-B replay	Show lab discovery map + null ablations	Score *your* paste
Temporal / Contra boxes	Only for **Map custom pair**	Affect Score
Map custom pair	Bake a **new** d′ from two pastes	Replace frozen discovery d
Example notes	Load short EX_TEMPORAL / EX_CONTRA	

Right panel — how to read Score

1. **Best layer** (green) — usually L24
2. **Bar chart** — mean |token·d| by layer (ramp L8→L24 is expected for temporal notes)
3. **Top sites** — tokenizer pieces with largest |score_d|
4. **Artifact path** — JSON under `artifacts/n2s-upstream-live-runs/`

Worked example — long FOLFOX note

Paste: metastatic CRC, FOLFOX response → later hepatic progression → discontinue (hedged with neuropathy).

Live Score (`LIVE_3c8c30b5`):

Signal	Value
Tokens	~737 note tokens · dirs cached
Best	**L24** |μ·d| ≈ **54**
Ramp	L8≈5 → L20≈29 → L24≈54
cos(d,v)	~0.03
Top toks	`atic` / `metast` / `pulmonary` / … (correlational)

Verdict: Score **worked**. Formation alignment is strong.
Not a clinical Dual decision — that is :8257 Evaluate.

Frozen replay after this? No — it ignores the paste.

Custom pair map

Builds `d′ = unit(μ_T′ − μ_C′)` from *your* two notes. Heavier (two HF captures). Still correlational. Use for teaching "map on my pair," not as the frozen discovery **d**.

Part IV — Code path (what powers the UI)

```
browser  static/upstream.html
     │   GET  /api/meta  /api/frozen  /api/job
     │   POST /api/score  /api/pair
     ▼
upstream_server.py          # ThreadingHTTPServer :8258
     ▼
upstream_api.py             # jobs, JSON shaping, claim_hygiene
```

```
    ▼
lab n2s_lab/
  n2s_upstream_live.py      # ensure_ua_directions, score_note_live, run_custom_pair_map
  n2s_upstream_prefill.py   # DEFAULT_NOTES, capture helpers
  hf_intervene.py           # prefill steers / ablate (lab U1/U-B; not Score buttons)
artifacts/
  n2s-upstream-ua-directions.pt
  n2s-upstream-ua-map.json / ub*.json
  n2s-upstream-live-runs/*.json
```

Bake / load d

```
# n2s_upstream_live.ensure_ua_directions
# If directions.pt exists → load cache.
# Else capture EX_TEMPORAL_FOLFOX + EX_CONTRA, then:
for L in (8, 12, 16, 20, 24):
    d = unit(mean(h_T[L]) - mean(h_C[L]))
torch.save(blob, "artifacts/n2s-upstream-ua-directions.pt")
```

Live Score

```
dirs = ensure_ua_directions(...)
captured = _capture_note(note)          # note-body residuals only
for layer, d in dirs.items():
    scores = residuals[layer] @ d       # each token
    # mean |scores|, top-k tokens, cos(d, freeze_v)
best_layer = argmax mean_abs_score
```

API sketch

```
# POST /api/score  → background job → score_note_live(...)
# GET  /api/job    → poll until result
# GET  /api/frozen → slim U-A map + U-B/U-B2 null tables (no HF)
# POST /api/pair   → run_custom_pair_map(temporal, contra)
```

Browser sketch

```
fetch('/api/score', { method:'POST', body: JSON.stringify({
  evidence: note.value, case_id, gold
})});
// poll /api/job → renderScore(bars + top sites)
```

Part V — How Upstream sits next to Downstream (do not mix claims)

	Upstream (:8258 / lab)	Downstream (:8257 / lab)
Object	**d** @ L24	**v** @ L20
Read	Score ·d	Deep dive
Write	α·d **null**	α·v **partial** (α=8)
Ablate	U-B null	Circuit C1 L20 MLP **causal site**
Sparse	U-C **locked off**	SAE pilot on **v**
Boundary	—	Track A Dual wrong_AUTO=0

Circuit C0–C2 (Downstream, frozen): top write L20/mlp along **v**; C1 soft_gate YES; C2 LOCAL_SUFFICIENT.
That does **not** reopen Upstream U-C/U-D.

Part VI — Claim hygiene

Say

- U-A formation map; strongest correlational sep @ **L24**
- cos(d, v) ≈ 0.03 — formation ≠ commit editor object
- Held-out paraphrase gen **soft+strong YES** (correlational)
- U-B / U-B2 / α·d **null** — no upstream editor
- `:8258` Score is a **correlational** live demo
- Track A contains uncertain transforms via REVIEW
- Downstream: α=8 partial editor + L20 MLP causal **site** (not full circuit)

Do not say

- Found the temporality neuron / full circuit
- Upstream editor / production clinical safety
- Score top tokens = mechanism
- Frozen replay scored my note
- Open U-C/U-D from nulls or from Downstream C0–C2
- α-chase expand-v; sealed-test redesign from these panels

Part VII — Reproduce / pointers

```
# UI
cd cxr-n2s-eval-workbench
../cxrlabs/faiss_gpu1/bin/python upstream_server.py

# Lab science
cd cxr-evidence-grounding-lab
./scripts/run_upstream_ua.sh map
./scripts/run_upstream_ub.sh
./scripts/run_upstream_ua_gen.sh panel
./scripts/run_upstream_ua_causal.sh panel --site commit

# Rebuild this PDF
cd cxr-n2s-eval-workbench
./build-upstream-tour-pdf.sh
```

Doc	Role
This file	Complete Upstream tour
`TRACKB-UPSTREAM-PORTFOLIO.md`	One-page portfolio
`TRACKB-UPSTREAM-PROGRAM.md`	Program + hard stop
`TRACKB-UPSTREAM-UA-GEN-FREEZE.md`	Gen freeze
`TRACKB-UPSTREAM-UA-CAUSAL.md`	α·d null
`TRACKB-UPSTREAM-PREFILL.md`	U0/U1
`N2S-SYSTEM-CLAIM.md`	System snapshot

Doc	Role
`N2S-Workbench-Newcomer-Tour.md`	Downstream twin
`TRACKB-CIRCUIT-C01-FREEZE.md`	Downstream circuit freeze

Closing line

Upstream: we can **see** and **generalize** a late-layer formation direction **d**, and Score real notes against it on `:8258` — but we could **not** turn that map into a causal prefill editor.

Downstream: we can **partially edit** commit utilization with **v** @ L20 and point to an L20 MLP **site**.

Two ports. Two objects (**d** vs **v**). Keep the math and the claims separate — and always decode `unit(μ_T − μ_C)` as: *unit difference of mean note-body residuals between temporal and contradiction discovery notes.*